

finest code. For instance, one may assess whether or not the request was correctly processed using the status code provided by `r.status_code` and `r.header`.

BeautifulSoup library

The *BeautifulSoup* module is used to extract information from the HTML and XML files. It offers a parse tree together with tools for exploring, searching, and updating it. It generates a parse tree from the page's source code, which is used to organise data more logically and hierarchically.

The purpose of this module is to perform rapid reversing tasks like screen scraping. It is effective due to three aspects.

1. It provides some simple and minimal codes for analysing a text and extracting the required items from the text. It provides facilities for directing, finding, and altering a parse tree.
2. It automatically converts outputs to UTF-8 and receives records at Unicode.
3. Parsers like LXML and HTML are often used to experiment with various parsing strategies or execution performances for flexibility.

```
pip install beautifulsoup4
```

Since this module is not also shipped with core Python it is necessary to install it with *pip* separately.

Examining the target website

Inspecting the target website is required before developing a

scripting program. One can do this by right-clicking on the target page and selecting *Developer tools*.

After clicking the *Developer tools* button of the open browser, one can inspect the internal codes of the open web page. We shall consider Chrome as the browser for this article.

The developer's tools allow a view of the site's document object model (DOM). From the display, one can examine the structure of the HTML coding of the page.

Parsing the HTML

After examining the scripting of the page, one can parse the raw HTML code into some useful information. For that, it's necessary to create a *BeautifulSoup* object by specifying the required parser. As this module is built on top of the HTML parsing libraries like *html5lib*, *lxml*, and *html.parser*, etc, all these libraries can be called along with this module. With this module, it's possible to parse a requested web page into its original HTML page. Here is an example:

```
import requests
from bs4 import BeautifulSoup

# GET request
r = requests.get('https://flask-bulma-css.appseed.us/')
# check the status code for the response received
print(r.status_code)

# Parsing the HTML
```

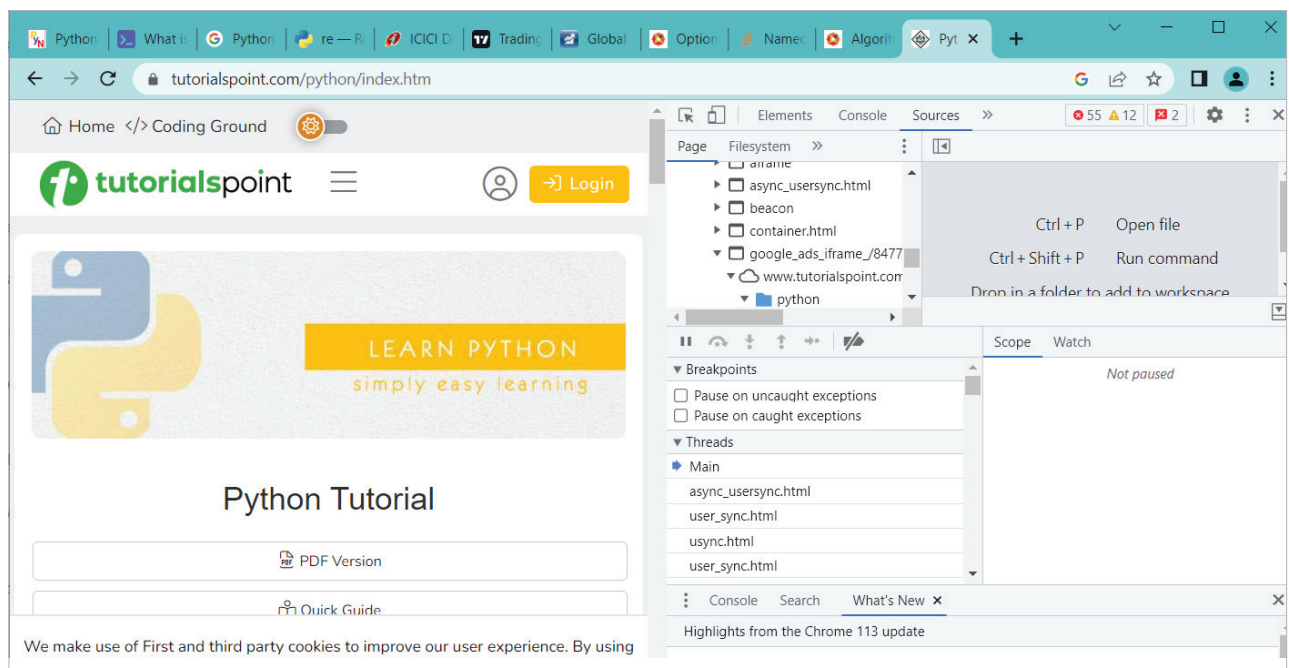


Figure 3: *Developer tool* in Google Chrome